FREIE UNIVERSITÄT BERLIN
DEPARTMENT OF BIOLOGY, CHEMISTRY, PHARMACY
INSTITUTE OF CHEMISTRY AND BIOCHEMISTRY

Physical & Theoretical Chemistry
Research Internship Report

A Systematic Evaluation of Data Acquisition Strategies for Machine Learning-driven Molecular and Material Property Prediction

Submitted by
Minh Nghia NGUYEN
5570279

Supervisors
Dr. Akhil Sugathtan NAIR
Prof. Dr. Beate PAULUS

Berlin, July, 2026

Statement of Academic Honesty

I hereby declare that I am the sole author of this seminar paper, that I have used no resources or aids other than the ones explicitly deemed permissible beforehand or explicitly cited in my text, that I have properly marked all ideas and/or passages taken from the works of others (including internet sources) as quotations and transparently cited their respective sources, that I have not submitted this paper or substantial portions of it for any other examination and will not do so in the future.

I understand that I have failed this examination if I do not meet the deadline communicated to me in advance, that I have failed this examination if I cheat or attempt to cheat, that serious deception may lead to my failing the entire module and may mean that I will not be allowed to continue studying this subject or discipline at German universities.

Berlin, July, 2026

Acknowledgments

I would like to express my sincere gratitude to the individuals and institutions who supported and contributed to this work during my internship.

First and foremost, I wish to express my deepest appreciation to my supervisor, Dr. Akhil Sugathan Nair, whose mentorship was invaluable. He generously shared his expertise, guided me through each stage of the research, and introduced me to the methods and computational tools essential to this work.

I am also truly grateful to Prof. Dr. Beate Paulus for welcoming me into her research group and for providing an environment for me to work in this field.

This project marked my first in-depth research experience at the intersection of machine learning and chemistry—specifically exploring active learning strategies to eliminate data redundancy across materials and molecular datasets.

Finally, I would like to thank my family for their unwavering encouragement throughout this journey, with a special thanks to my father and brother for their insightful guidance in helping me navigate machine learning concepts.

List of Abbreviations and Acronyms

AL	Active Learning
DFT	Density Functional Theory
GNN	Graph Neural Network
GCN	Graph Convolutional Network
KRR	Kernel Ridge Regressor
MAE	Mean Average Error
ML	Machine Learning
MSE	Mean Square Error
RFR	Random Forest Regressor
RMSE	Root Mean Square Error
XGB	Extreme Gradient Boost

List of Figures

List of Tables

Contents

1 Introduction **7**

1.1 Background and Motivation 7

2 Theoretical Background **7**

3 Results and Discussion **7**

4 Conclusion **7**

5 References **7**

1 Introduction

1.1 Background and Motivation

2 Theoretical Background

3 Results and Discussion

4 Conclusion

5 References